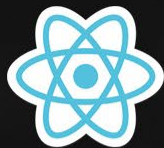




React hands on workshop

By Vijay Shivakumar



Welcome to the React workshop



What do we need before we begin... ?

Technical Skill : HTML5, CSS3, JavaScript 1.8.5

Hardware and software:

IDE : visual studio code

Browsers : chrome latest

Platform : nodejs latest

Database : mongodb

Version Control : git

Network : internet access to download from git and npmjs.org



Objectives

- Understand and explore ES6 / ES7
- Write Programs using Pure React
- Understand JSX usage
- Develop programs using React platform
- Workflow with Context API
- Workflow with Redux
- Usage of middleware Saga
- Unit testing with Jest



What are we learning in this course ?

ES6+

Functional programming

Arrow functions

Immutable objects

Template strings

Destructuring

Array Methods

Scope Management

History API

CSS / SASS

What is React ?

Tooling and setup for

React-CLI

React Components

Understanding JSX

Data binding

Class and Style binding

State

Props and PropTypes

Conditional Rendering

Working with Forms

Events

Context API

Lifecycle Methods

Working with HTTP

Provider API

Redux

React Routing

Lists and Keys

Fragments

Firebase / MongoDB



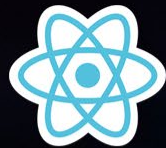
Your Instructor / About Me

Vijay Shivakumar
Designer | Developer | Trainer



Certified Expert

Training & Consultation of
Contemporary Web Technologies and Adobe products from past 18+ years



Audience / About you

Developer
Designer
Manager
Architect
Technology Enthusiast



Prerequisites / Before we begin

HTML 5

CSS3

ES6

NodeJS

TypeScript

Webpack

Express

Babel

TDD

MongoDB



BABEL





Must Know / Before we begin

HTML 5
CSS3

I assume you know



ES6

ES6

block scope
de structuring
arrow function
default parameters
spread operator

array methods
template strings
classes
modules



Must Know / Before we begin

HTML 5
CSS3

I assume you know



ES6

We shall learn these

ES6

block scope
de structuring
default parameters
rest params
spread syntax

arrow function
array methods
template strings
classes
modules



About React / Why do we need react ?

Client Side Programming LIBRARY

Created and maintained by Facebook developers

Used to build dynamic user interfaces (Frontend)

Everything is a component

Often referred as V in the MVC



What makes React great / Principles of React

DOM Manipulation only with React

Components architecture / Composability

One way data flow (Unidirectional Data Flow)

A solid UI library



Why React / Advantages of React

Component architecture

Easy to scale existing applications one component at a time

Partial refresh of UI Virtual DOM

Only renders the area that is modified avoiding page refresh

Fast

Client get a faster response from application and they are happy

Leverage on ES6 and later

Existing knowledge of JavaScript can be used to scale with react



Benefits / Key take away from this course

Components

- Just like functions

- Reusable and composable

- Can manage a private state

Reactive updates

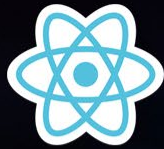
- Updates with user interaction

- Take updates to the browser

Virtual view in memory

- write HTML in JavaScript

- Tree reconciliation



React Architecture

APP (component)

HEADER (component)

MAIN (component)

ARTICLE (component)

ARTICLE (component)

**HEADER
(component)**

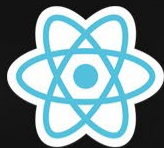
**ASIDE
(component)**

FOOTER (component)



JSX / Rules

1. JSX must return a single root element or a component
2. Orphan / self closing tags needs to be closed eg `<br/>` `<hr/>` `<img/>` (don't use `<br>`)
3. If you need to return multiple element and don't want to wrap them in a div use `React.Fragment` or `<>`;
4. Interpolation using `{ 2 + 3 }`
5. use `className` instead of `class` attribute
6. `htmlFor` instead of 'for' attribute on label elements
7. use `defaultValue` instead of `value` on input element to create uncontrolled inputs
8. use camel cased so refer type for events for eg, `onClick`, `onMouseOver`
9. while using inline style use a config object
10. style properties should be camel cased
11. HTML elements name must be lowercase like `h1`, `div`, `img`, `input`, `button` etc
12. jsx classes, function names must begin with uppercase characters like `Panel`, `UserForm`, `Datagrid` etc



Components



Components / Building blocks of an application

Composition

MAIN (component)

ARTICLE (component)

ARTICLE (component)

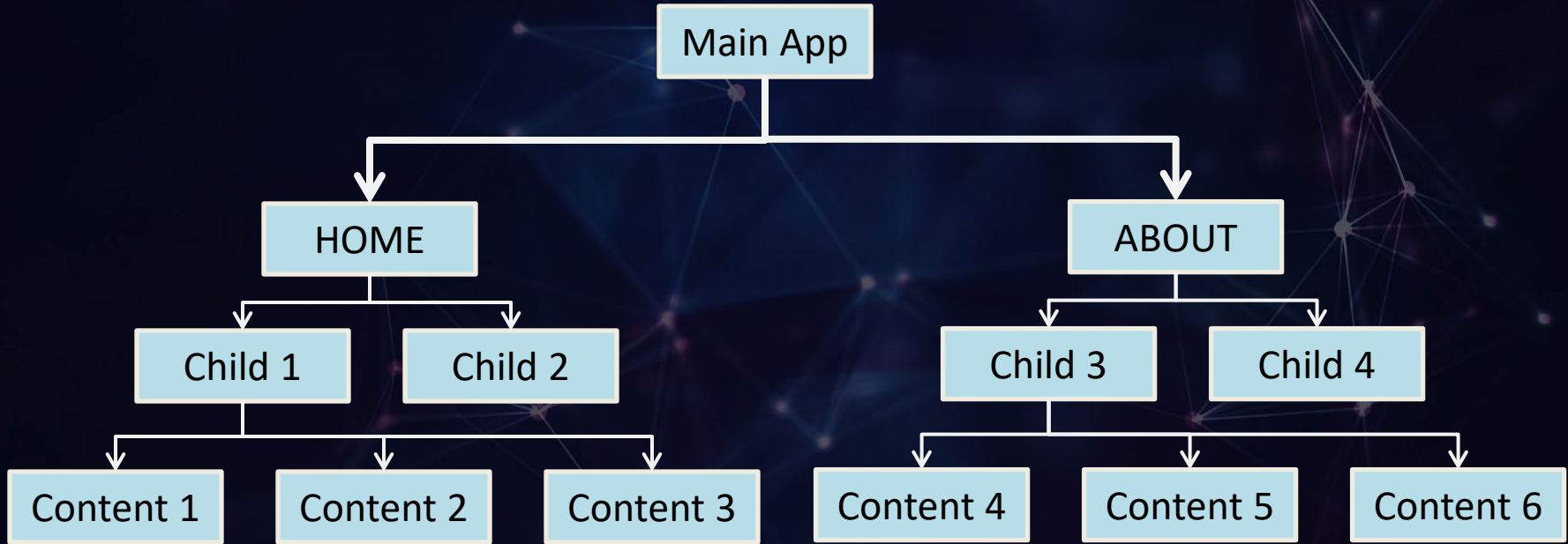


Component / Types

Function components are stateless hence less dynamic
Class components are statefull and are dynamic

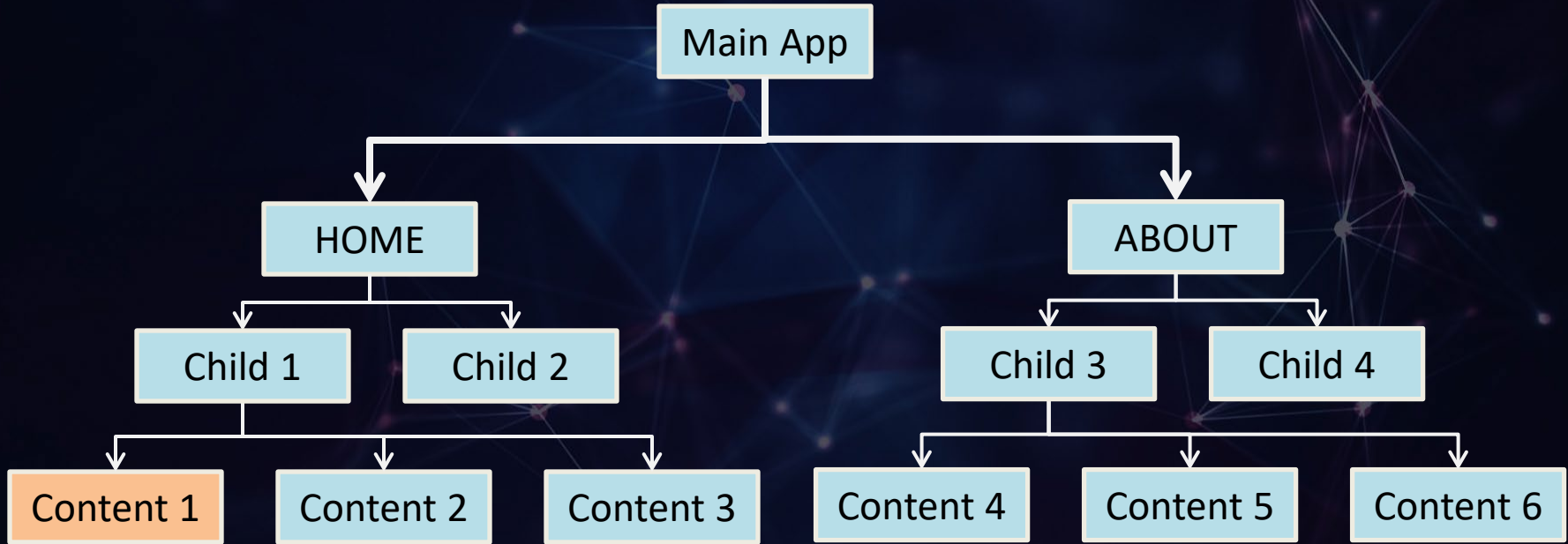


Data flow issues



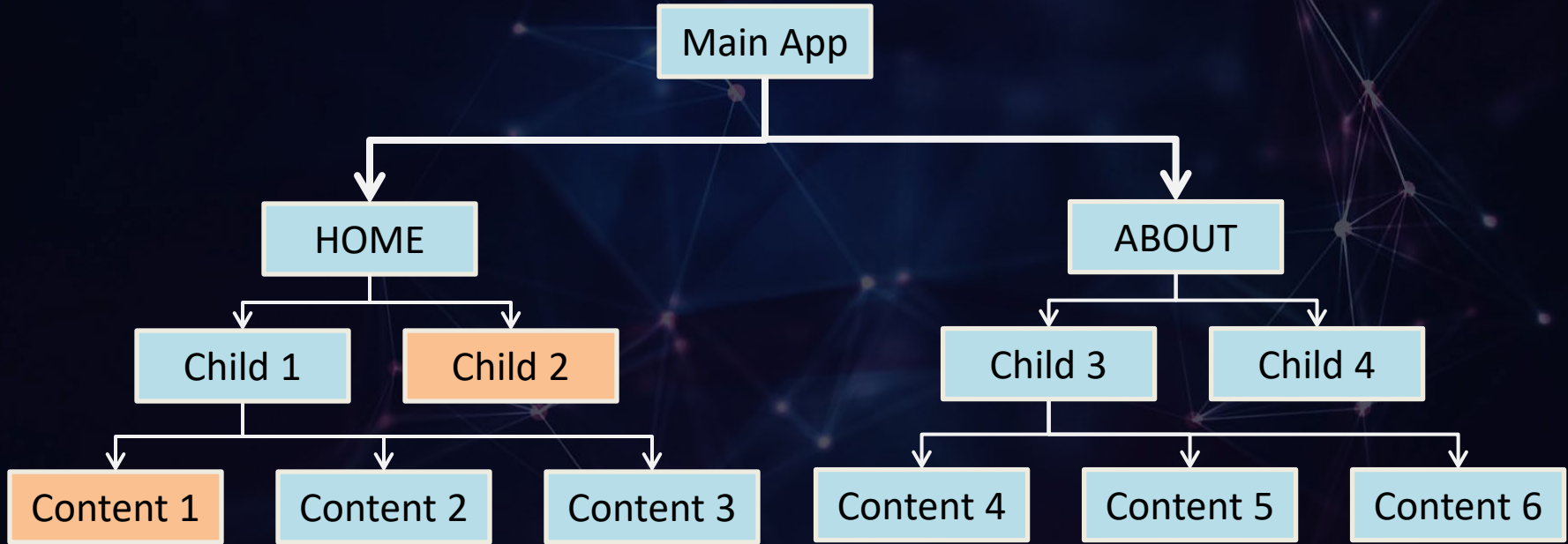


Data flow issues



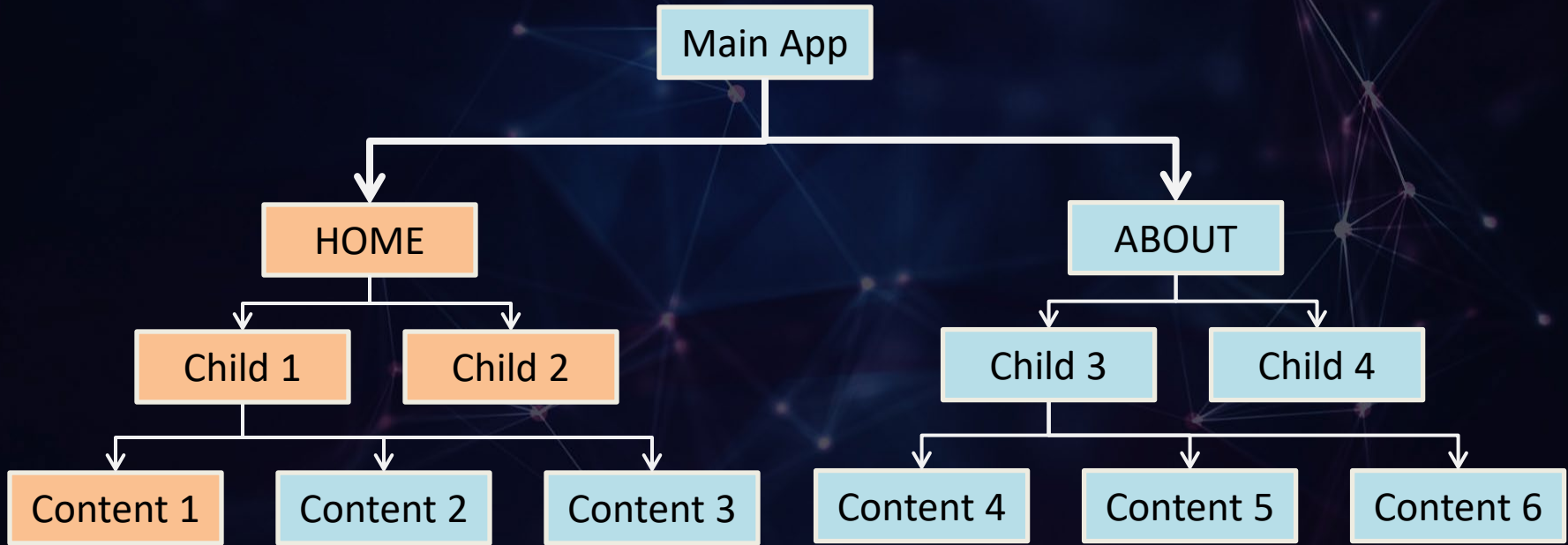


Data flow issues



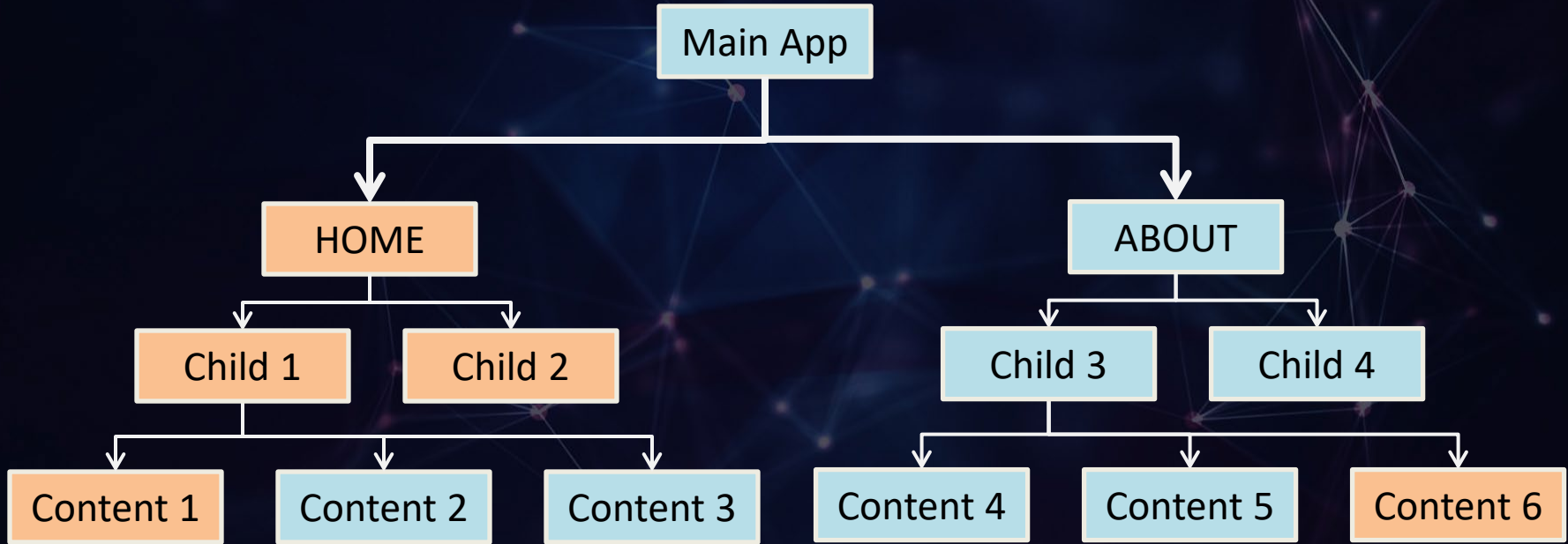


Data flow issues



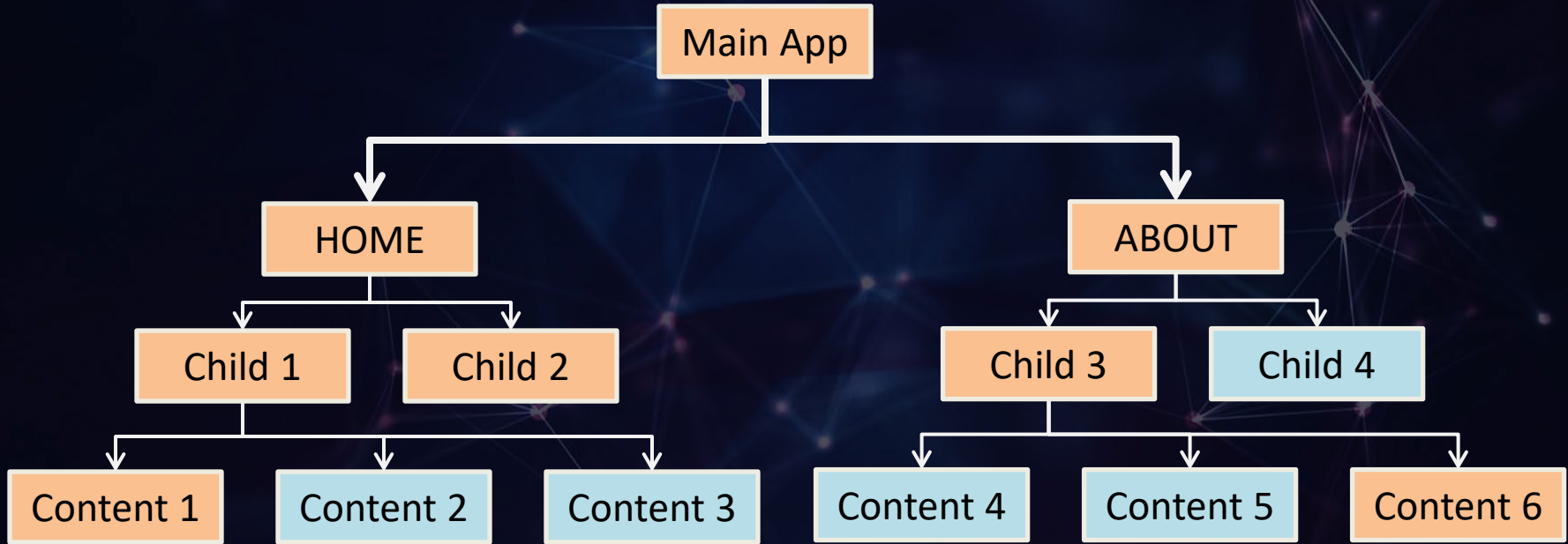


Data flow issues



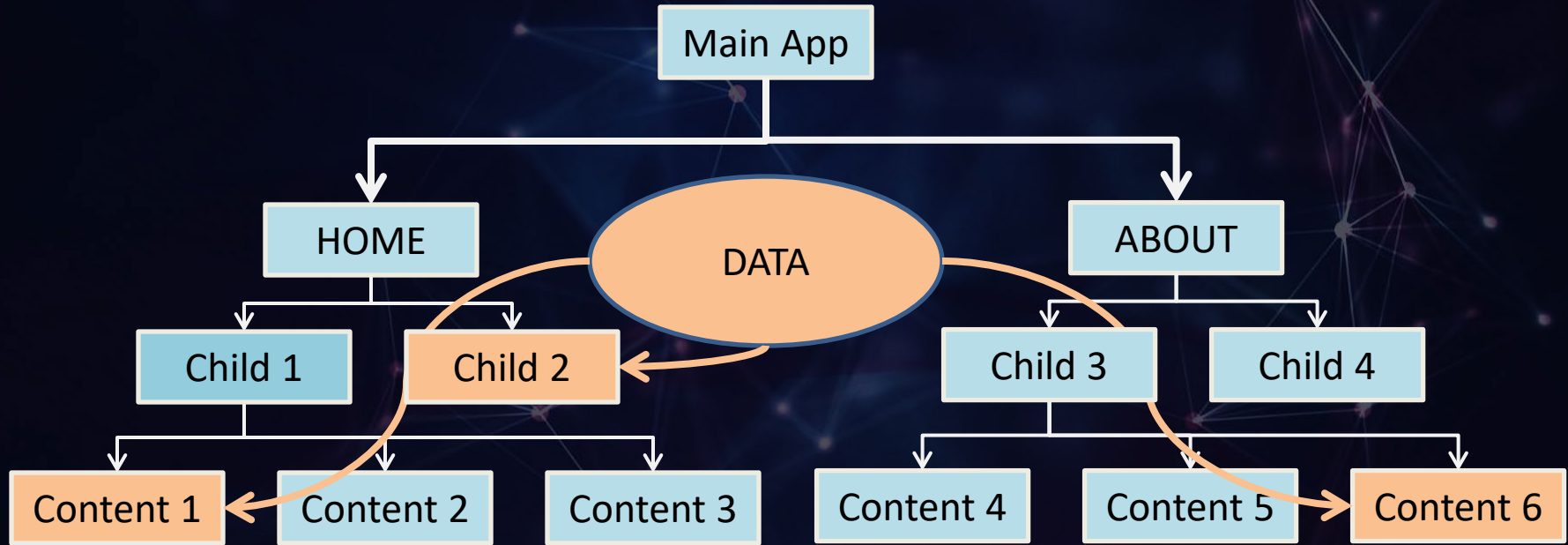


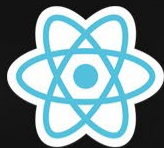
Data flow issues





Data flow issues





Redux



What is Redux ?

Manages Data Store that can be accessed across your app

Redux makes state management more predictable by having a single source or truth (single place to store the state)

We can set strict rules for how the state can be updated (via reducers)



What is Redux ?

A store — an immutable object that holds the application's state data

A reducer — a function that returns state data, triggered by an action type

An action — an object that tells the reducer how to change the state. It must contain a type property, and it can contain an optional payload property



Three Principles of Redux

Application's state is stored in a single object tree which is managed by redux store.

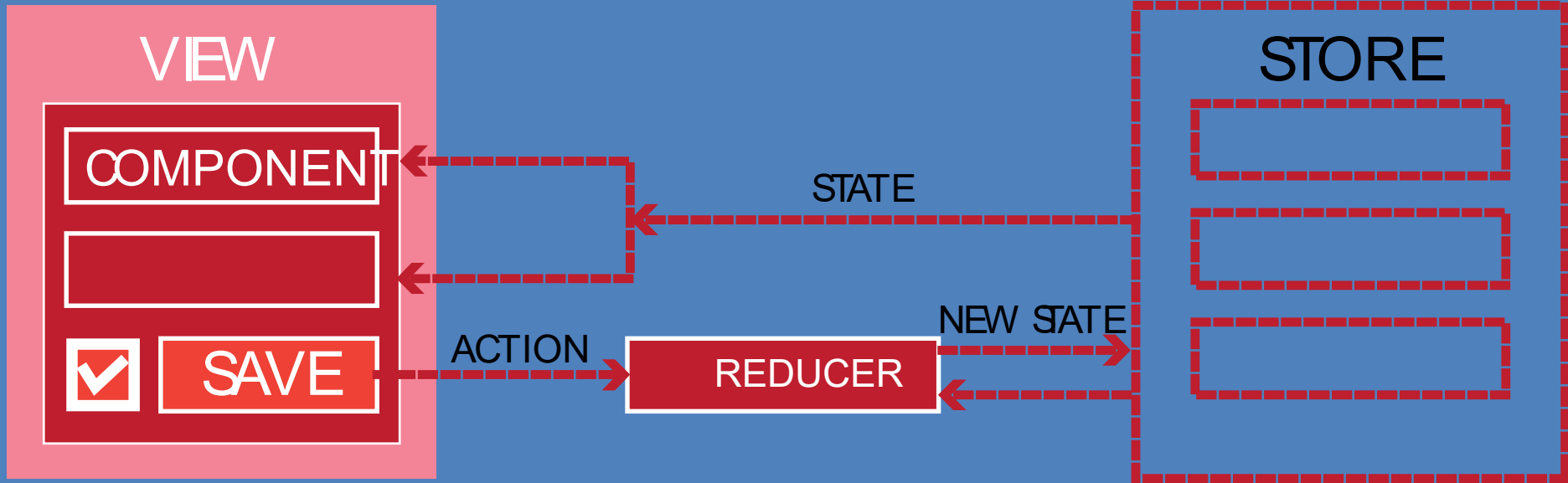
Only way to change the state is via a reducer function which knows what type of action happened.

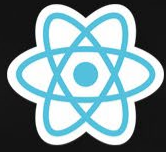
(should not update the state by any other means)

How the state tree is transformed by action that are directed by reducer function



Redux Architecture





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